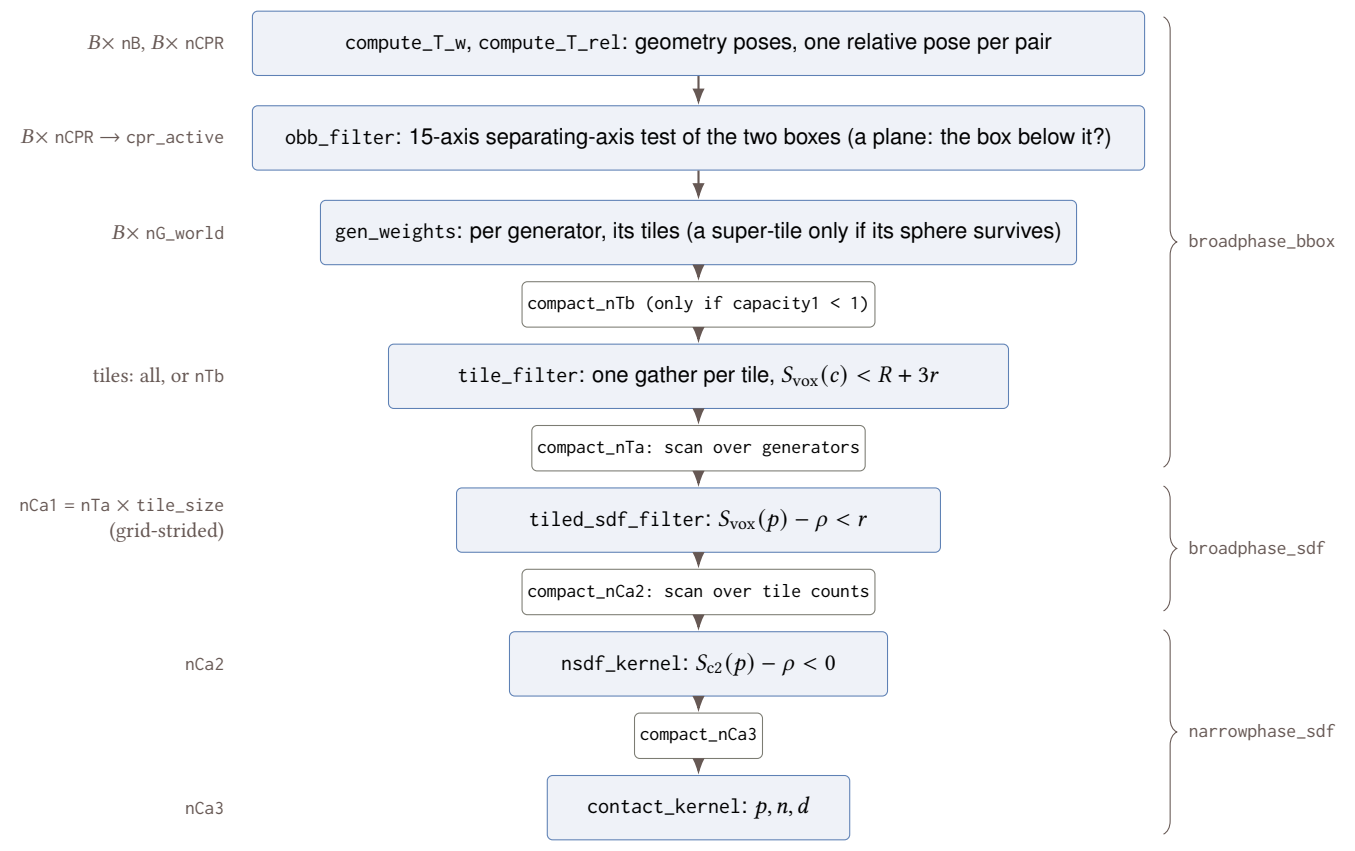

$B\times$ nB, $B\times$ nCPR
compute_T_w, compute_T_rel: geometry poses, one relative pose per pair
$B\times$ nCPR → cpr_active
obb_filter: 15-axis separating-axis test of the two boxes (a plane: the box below it?)
$B\times$ nG_world
gen_weights: per generator, its tiles (a super-tile only if its sphere survives)
compact_nTb (only if capacity1 < 1)
tiles: all, or nTb
tile_filter: one gather per tile, $S_{\text{vox}}(c) < R + 3r$
compact_nTa: scan over generators
broadphase_bbox
nCa1 = nTa × tile_size (grid-strided)
tiled_sdf_filter: $S_{\text{vox}}(p) - \rho < r$
compact_nCa2: scan over tile counts
broadphase_sdf
nCa2
nsdf_kernel: $S_{c2}(p) - \rho < 0$
compact_nCa3
nCa3
contact_kernel: p, n, d
narrowphase_sdf